## 4.8 Exercises

### Exercise 4.1: Input-to-state stability and convergence

Assume the nonlinear system

$$x^+ = f(x, u)$$

is input-to-state stable (ISS) so that for all $x_0 \in \mathbb{R}^n$, input sequences $\mathbf{u}$, and $k \geq 0$

$$|x(k; x_0, \mathbf{u})| \leq \beta(|x_0|, k) + \gamma(\|\mathbf{u}\|)$$

in which $x(k; x_0, \mathbf{u})$ is the solution to the system equation at time $k$ starting at state $x_0$ using input sequence $\mathbf{u}$, and $\gamma \in \mathcal{K}$ and $\beta \in \mathcal{KL}$.

(a) Show that the ISS property also implies

$$|x(k; x_0, \mathbf{u})| \leq \beta(|x_0|, k) + \gamma(\|\mathbf{u}\|_{0:k-1})$$

in which $\|\mathbf{u}\|_{a:b} = \max_{a \leq j \leq b} |u(j)|$.

(b) Show that the ISS property implies the "converging-input converging-state" property (Jiang and Wang, 2001), (Sontag, 1998, p. 330), i.e., show that if the system is ISS, then $u(k) \to 0$ implies $x(k) \to 0$.

### Exercise 4.2: Output-to-state stability and convergence

Assume the nonlinear system

$$x^+ = f(x) \qquad y = h(x)$$

is output-to-state stable (OSS) so that for all $x_0 \in \mathbb{R}^n$ and $k \geq 0$

$$|x(k; x_0)| \leq \beta(|x_0|, k) + \gamma(\|\mathbf{y}\|_{0:k})$$

in which $x(k; x_0)$ is the solution to the system equation at time $k$ starting at state $x_0$, and $\gamma \in \mathcal{K}$ and $\beta \in \mathcal{KL}$.

Show that the OSS property implies the "converging-output converging-state" property (Sontag and Wang, 1997, p. 281) i.e., show that if the system is OSS, then $y(k) \to 0$ implies $x(k) \to 0$.

### Exercise 4.3: i-IOSS and convergence

Prove Proposition 4.2, which states that if system

$$x^+ = f(x, w) \qquad y = g(x)$$

is i-IOSS, and $w_1(k) \to w_2(k)$ and $y_1(k) \to y_2(k)$ as $k \to \infty$, then

$$x(k; z_1, \mathbf{w}_1) \to x(k; z_2, \mathbf{w}_2) \quad \text{as } k \to \infty \quad \text{for all } z_1, z_2$$

### Exercise 4.4: Observability and detectability of linear time-invariant systems and OSS

Consider the linear time-invariant system

$$x^+ = Ax \qquad y = Cx$$

(a) Show that if the system is observable, then the system is OSS.

(b) Show that the system is detectable if and only if the system is OSS.

### Exercise 4.5: Observability and detectability of linear time-invariant system and IOSS

Consider the linear time-invariant system with input

$$x^+ = Ax + Gw \qquad y = Cx$$

(a) Show that if the system is observable, then the system is IOSS.

(b) Show that the system is detectable if and only if the system is IOSS.

### Exercise 4.6: Max or sum?

To facilitate complicated arguments involving $\mathcal{K}$ and $\mathcal{KL}$ functions, it is often convenient to interchange sum and max operations. First some suggestive notation: let the max operator over scalars be denoted with the $\oplus$ symbol so that

$$a \oplus b := \max(a, b)$$

(a) Show that the $\oplus$ operator is commutative and associative, i.e., $a \oplus b = b \oplus a$ and $(a \oplus b) \oplus c = a \oplus (b \oplus c)$ for all $a, b, c$, so that the following operation is well defined and the order of operation is inconsequential

$$a_1 \oplus a_2 \oplus a_3 \cdots \oplus a_n := \bigoplus_{i=1}^{n} a_i$$

(b) Find scalars $d$ and $e$ such that for all $a, b \geq 0$, the following holds

$$d(a + b) \leq a \oplus b \leq e(a + b)$$

(c) Find scalars $\overline{d}$ and $\overline{e}$ such that for all $a, b \geq 0$, the following holds

$$\overline{d}(a \oplus b) \leq a + b \leq \overline{e}(a \oplus b)$$

(d) Generalize the previous result to the $n$ term sum; find $d_n, e_n, \overline{d}_n, \overline{e}_n$ such that the following holds for all $a_i \geq 0$, $i = 1, 2, \ldots, n$

$$d_n \sum_{i=1}^{n} a_i \leq \bigoplus_{i=1}^{n} a_i \leq e_n \sum_{i=1}^{n} a_i$$
$$\overline{d}_n \bigoplus_{i=1}^{n} a_i \leq \sum_{i=1}^{n} a_i \leq \overline{e}_n \bigoplus_{i=1}^{n} a_i$$

(e) Show that establishing convergence (divergence) in sum or max is equivalent, i.e., consider a time sequence $(s(k))_{k \geq 0}$

$$s(k) = \sum_{i=1}^{n} a_i(k) \qquad \overline{s}(k) = \bigoplus_{i=1}^{n} a_i(k)$$

Show that

$$\lim_{k \to \infty} s(k) = 0 \ (\infty) \text{ if and only if } \lim_{k \to \infty} \overline{s}(k) = 0 \ (\infty)$$

### Exercise 4.7: Where did my constants go?

Once $\mathcal{K}$ and $\mathcal{KL}$ functions appear, we may save some algebra by switching from the sum to the max.

In the following, let $\gamma(\cdot)$ be any $\mathcal{K}$ function.

(a) If you choose to work with sum, derive the following bounding inequalities (Rawlings and Ji, 2012)

$$\gamma(a_1 + a_2 + \cdots + a_n) \leq \gamma(na_1) + \gamma(na_2) + \cdots + \gamma(na_n)$$
$$\gamma(a_1 + a_2 + \cdots + a_n) \geq \frac{1}{n}\Big(\gamma(a_1) + \gamma(a_2) + \cdots + \gamma(a_n)\Big)$$

(b) If you choose to work with max instead, derive instead the following simpler result

$$\gamma(a_1 \oplus a_2 \oplus \cdots \oplus a_n) = \gamma(a_1) \oplus \gamma(a_2) \oplus \cdots \oplus \gamma(a_n)$$

Notice that you have an equality rather than an inequality, which leads to tighter bounds.

### Exercise 4.8: Linear systems and incremental stability

Show that for a linear time-invariant system, i-ISS (i-OSS, i-IOSS) is equivalent to ISS (OSS, IOSS).

### Exercise 4.9: Nonlinear observability and Lipschitz continuity implies i-OSS

Consider the following definition of observability for nonlinear systems in which $f$ and $h$ are Lipschitz continuous. A system

$$x^+ = f(x) \qquad y = h(x)$$

is observable if there exists $N_0 \in \mathbb{I}_{\geq 1}$ and $\mathcal{K}$ function $\gamma$ such that

$$\sum_{k=0}^{N_0-1} \left| y(k;x_1) - y(k;x_2) \right| \geq \gamma(|x_1 - x_2|) \tag{4.63}$$

holds for all $x_1, x_2 \in \mathbb{R}^n$. This definition was used by Rao et al. (2003) in showing stability of nonlinear MHE to initial condition error under zero state and measurement disturbances.

(a) Show that this form of nonlinear observability implies i-OSS.

(b) Show that i-OSS does not imply this form of nonlinear observability and, therefore, i-OSS is a weaker assumption.

The i-OSS concept generalizes the linear system concept of detectability to nonlinear systems.

### Exercise 4.10: Equivalance of detectability and IOSS for continuous time, linear, time-invariant system

Consider the continuous time, linear, time-invariant system with input

$$\dot{x} = Ax + Bu \qquad y = Cx$$

Show that the system is detectable if and only if the system is IOSS.

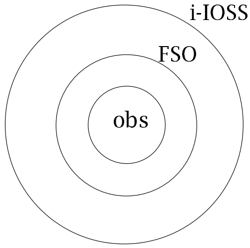


**Figure 4.11:** Relationships between i-IOSS, FSO, and observable for $\mathcal{K}$-continuous nonlinear systems.

### Exercise 4.11: Observable, FSO, and detectable for linear systems

Consider the linear time-invariant system

$$x^+ = Ax \qquad y = Cx$$

and its observability canonical form. What conditions must the system satisfy to be

(a) observable?

(b) final-state observable (FSO)?

(c) detectable?

### Exercise 4.12: Robustly GAS implies GAS in estimation

Show that an RGAS estimator is also GAS.

### Exercise 4.13: Relationships between observable, FSO, and i-IOSS

Show that for the nonlinear system $x^+ = f(x, w)$, $y = h(x)$ with globally $\mathcal{K}$-continuous $f(\cdot)$ and $h(\cdot)$, i-IOSS, FSO, and observable are related as shown in the Venn diagram in Figure 4.11.

### Exercise 4.14: Observabilty plus $\mathcal{K}$-continuity imply FSO

Prove Proposition 4.16. Hint: first try replacing global $\mathcal{K}$-continuity with the stronger assumption of global Lipschitz continuity to get a feel for the argument.

### Exercise 4.15: Detectable linear time-invariant system and i-UIOSS

Show that the detectable linear time-invariant system $x^+ = Ax + Bu + Gw$, $y = Cx$ is i-UIOSS from Definition 4.42.

### Exercise 4.16: Dynamic programming recursion for Kalman predictor

In the Kalman predictor, we use forward DP to solve at stage $k$

$$\min_{x,w} \ell(x, w) + V_k^-(x) \qquad \text{s.t. } z = Ax + w$$

in which $x$ is the state at the current stage and $z$ is the state at the next stage. The stage cost and arrival cost are given by

$$\ell(x,w) = (1/2)\big( \left| y(k) - Cx \right|^2_{R^{-1}} + w'Q^{-1}w \big) \qquad V_k^-(x) = (1/2)\left| x - \hat{x}^-(k) \right|^2_{(P^-(k))^{-1}}$$

and we wish to find the value function $V^0(z)$, which we denote $V_{k+1}^-(z)$ in the Kalman predictor estimation problem.

(a) Combine the two $x$ terms to obtain

$$\min_{x,w} \frac{1}{2}\Big( w'Q^{-1}w + (x - \hat{x}(k))'P(k)^{-1}(x - \hat{x}(k)) \Big) \qquad \text{s.t. } z = Ax + w$$

and, using the third part of Example 1.1, show

$$\begin{aligned} P(k) &= P^-(k) - P^-(k)C'(CP^-(k)C' + R)^{-1}CP^-(k) \\ L(k) &= P^-(k)C'(CP^-(k)C' + R)^{-1}C'R^{-1} \\ \hat{x}(k) &= \hat{x}^-(k) + L(k)(y(k) - C\hat{x}^-(k)) \end{aligned}$$

(b) Add the $w$ term and use the inverse form in Exercise 1.18 to show the optimal cost is given by

$$\begin{aligned} V^0(z) &= (1/2)(z - A\hat{x}^-(k+1))'(P^-(k+1))^{-1}(z - A\hat{x}^-(k+1)) \\ \hat{x}^-(k+1) &= A\hat{x}(k) \\ P^-(k+1) &= AP(k)A' + Q \end{aligned}$$

Substitute the results for $\hat{x}(k)$ and $P(k)$ above and show

$$\begin{aligned} V_{k+1}^-(z) &= (1/2)(z - \hat{x}^-(k+1))'(P^-(k+1))^{-1}(z - \hat{x}(k+1)) \\ P^-(k+1) &= Q + AP^-(k)A' - AP^-(k)C'(CP^-(k)C' + R)^{-1}CP^-(k)A \\ \hat{x}^-(k+1) &= A\hat{x}^-(k) + \tilde{L}(k)(y(k) - C\hat{x}^-(k)) \\ \tilde{L}(k) &= AP^-(k)C'(CP^-(k)C' + R)^{-1} \end{aligned}$$

(c) Compare and contrast this form of the estimation problem to the one given in Exercise 1.29 that describes the Kalman filter.

### Exercise 4.17: Duality, cost to go, and covariance

Using the duality variables of Table 4.2, translate Theorem 4.12 into the version that is relevant to the state estimation problem.

### Exercise 4.18: Estimator convergence for $(A, G)$ not stabilizable

What happens to the stability of the optimal estimator if we violate the condition

$$(A, G) \text{ stabilizable}$$

(a) Is the steady-state Kalman filter a stable estimator? Is the full information estimator a stable estimator? Are these two answers contradictory? Work out the results for the case $A = 1, G = 0, C = 1, P^-(0) = 1, Q = 1, R = 1$.
Hint: you may want to consult de Souza, Gevers, and Goodwin (1986).

(b) Can this phenomenon happen in the LQ regulator? Provide the interpretation of the time-varying regulator that corresponds to the time-varying filter given above. Does this make sense as a regulation problem?

### Exercise 4.19: Exponential stability of the Kalman predictor

Establish that the Kalman predictor defined in Section 4.2.1 is a globally exponentially stable estimator. What is the corresponding linear quadratic regulator?

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